ELTPO501 – POWER PLANT OPERATION
DATE: …/…/2025

PERIOD: ……

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

SECTOR: Energy

TRADE NAME: Electrical Technology

RQF LEVEL: 5

MODULE TITLE: – POWER PLANT OPERATION

MARKS: **……/100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Twenty-four (24) questions distributed in Three Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: Between (11-17) questions, all are **Compulsory**. **55 marks**
Section **B**: Five (5) questions, choose and answer any three (3). **30 marks**
Section **C**: Two (2) questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

Section **A**: Between (11-17) questions, all are **Compulsory**. **55 marks**

MULTIPLE-CHOICE QUESTIONS (20 MARKS)

1. Which of the following best defines a power plant?
A. A building used to store mechanical equipment
B. A facility that generates electricity for distribution
C. A system that only uses renewable resources
D. A station used for water purification

2. Which of the following is a renewable source of energy?
A. Coal
B. Natural gas
C. Solar energy
D. Petroleum

3. Which of the following is a major disadvantage of hydroelectric power plants?
A. They cause heavy smoke emissions
B. High capital cost due to dam construction
C. They require a large amount of fuel
D. They cannot start quickly

4. In a hydropower plant, the penstock is used to:
A. Filter impurities in water
B. Carry water from the reservoir to the turbine
C. Store water during dry seasons
D. Convert mechanical energy into electricity

5. Which turbine is commonly used for high-head hydropower plants?
A. Kaplan
B. Francis
C. Pelton wheel
D. Gas turbine

6. Which type of boiler is most widely used in large steam power stations?
A. Fire tube boiler
B. Water tube boiler
C. Horizontal tubular boiler
D. Locomotive boiler

7. In a nuclear reactor, the function of the moderator is to:
A. Absorb excess neutrons
B. Increase the temperature of coolant
C. Slow down fast-moving neutrons
D. Convert heat into steam

8. In a nuclear power plant, the control rods are made of:
A. Lead
B. Sodium
C. Cadmium
D. Thorium

9. In a gas turbine plant, the starting motor is used to:
A. Rotate the turbine at full speed
B. Start the compressor before turbine operation
C. Preheat the combustion chamber
D. Remove exhaust gases

10. The overall efficiency of a typical steam power station is about:
A. 10%
B. 20%
C. 29–30%
D. 50%

TRUE OR FALSE QUESTIONS (20 MARKS)

1. More than 50% of the heat from coal combustion is lost in the condenser of a steam power plant

2. Superheated steam reduces blade corrosion in turbines.

3. A surface condenser mixes cooling water and steam directly.)

4. A nuclear power plant requires large amounts of coal supply.

5. Gas turbine power plants require very large amounts of water for their condensers.
6. Pelton wheel turbines are used for low-head applications.

7. Diesel power plants have high running costs due to expensive fuel.

8. Spillways are used to discharge excess water from the reservoir.

9. A surge tank stores fuel for the diesel power plant.

10. A hydroelectric power plant cannot be installed in hilly areas.

MATCHING QUESTIONS (5 MARKS)

Match the component in Column A with its correct description in Column B.

COLUMN A

1. Penstock
2. Alternator
3. Spillway
4. Boiler
5. Diesel Engine

COLUMN B

A. Converts mechanical energy into electrical energy
B. Carries water under pressure to the turbine
C. Burns fuel to generate mechanical power
D. Releases excess water from the dam
E. Produces steam by burning coal

FILL-IN-THE-BLANK QUESTIONS (10 MARKS)

1. The _____ cycle is used in steam power plants to convert heat to mechanical energy.

2. In a hydropower system, the _____ converts hydraulic energy into mechanical energy.

3. The device used to reduce pressure swings in the penstock is called a _____.

4. Coal, oil, and natural gas are examples of _____ energy sources.

5. A _____ power plant uses diesel fuel to generate electricity**.**

Section **B**: Five (5) questions, choose and answer any three (3). **30 marks**

1. Explain the difference between renewable and non-renewable energy sources, giving two examples of each.

2. Describe how a hydroelectric power plant converts stored water into electrical energy

can they be minimized?

3. Identify the major auxiliary systems in a diesel power plant and explain their functions.

4. Discuss the technical, economic, and environmental factors that influence the choice of a power plant type for a specific region.

5. What are the major factors that must be considered when selecting a site for a nuclear power station?

Section **C**: Two (2) questions, choose and answer any one (1) **15 marks**

1. Describe, step by step, how energy is converted from nuclear fission into electrical energy in a nuclear power plant.
2. Fill the name of the main components of a hydropower plant

